

Maps and Isomorphisms

Functions, algebraic structure, and smooth maps

The properties of a map depend on its domain, its codomain, and the structure that it preserves. Injectivity and surjectivity determine whether a map has an inverse as a function. An isomorphism must also preserve the relevant operations. We develop these notions and their proof methods, then apply them to the identification of vectors in $\mathbb{R}^n$ with derivations at a point.

1 Maps, images, and preimages

Definition 1.1: Function, domain, and codomain

A function $F : A \rightarrow B$ assigns to each $a \in A$ exactly one element $F(a) \in B$. The set A is the *domain*, and B is the *codomain*. The *image* is the subset

$$F(A) = \{F(a) : a \in A\} \subseteq B.$$

For $b \in B$, the set $F^{-1}(\{b\}) = \{a \in A : F(a) = b\}$ consists of the preimages of b . This notation for a preimage set does not require an inverse function.

A formula alone does not determine the properties of a map. For example, $x \mapsto x^2$ behaves differently as a map $\mathbb{R} \rightarrow \mathbb{R}$, as a map $[0, \infty) \rightarrow \mathbb{R}$, and as a map $[0, \infty) \rightarrow [0, \infty)$. The domain and codomain must therefore be specified as part of the map.

Logical forms

Injective	For all $a, a' \in A$, $F(a) = F(a')$ implies $a = a'$.
Surjective	For every $b \in B$, there exists $a \in A$ with $F(a) = b$.
Bijective	Both statements hold.

The quantifiers determine the initial step of a proof. Injectivity concerns two arbitrary elements of the domain. Surjectivity concerns an arbitrary element of the codomain and the existence of a preimage.

Maps on equivalence classes

If an input is an equivalence class, a formula involving a representative might give different outputs for different representatives. Before studying injectivity or surjectivity, prove that the output is independent of the representative. For example, $[f]_p \mapsto f(p)$ is well defined on smooth function germs because functions that agree near p have the same value at p . This is a separate question from whether the map is injective.

2 Injective maps

Definition 2.1: Injective map

A map $F : A \rightarrow B$ is *injective* if

$$F(a) = F(a') \implies a = a' \quad \text{for every } a, a' \in A.$$

Equivalently, no element of B has two distinct preimages in A .

Establishing injectivity

Take arbitrary $a, a' \in A$ and assume $F(a) = F(a')$. Use that equality and the definition of F to derive $a = a'$. You may instead assume $F(a) = F(a')$ and show that $a - a' = 0$ when A is a vector space. To disprove injectivity, find distinct a, a' with the same image.

Example 2.2: Restriction of the square map

The map $q : \mathbb{R} \rightarrow \mathbb{R}$, $q(x) = x^2$, is not injective because $q(1) = q(-1) = 1$ but $1 \neq -1$.

The restriction $q : [0, \infty) \rightarrow \mathbb{R}$ is injective. If $x, y \geq 0$ and $x^2 = y^2$, then $x = y$ because nonnegative numbers have a unique nonnegative square root. The conclusion depends on the domain.

Proposition 2.3: The kernel criterion

Let $T : V \rightarrow W$ be linear. Then T is injective if and only if $\ker T = \{0_V\}$, where $\ker T = \{v \in V : T(v) = 0_W\}$.

Proof. If T is injective and $T(v) = 0_W = T(0_V)$, then $v = 0_V$. Conversely, if $\ker T = \{0_V\}$ and $T(u) = T(v)$, linearity gives $T(u - v) = T(u) - T(v) = 0_W$. Hence $u - v = 0_V$, so $u = v$. $\square$

For a linear map, injectivity can therefore be established by fixing $v \in V$ and assuming $T(v) = 0$. If $T(v)$ is an operator, this equality means that it vanishes on every function in its domain. Suitable test functions may be chosen to recover the components of v .

3 Surjective maps

Definition 3.1: Surjective map

A map $F : A \rightarrow B$ is *surjective* if $F(A) = B$. Written with quantifiers, this says

for every $b \in B$ there exists $a \in A$ such that $F(a) = b$.

Different inputs may have the same output. Surjectivity requires at least one preimage for each target, not exactly one.

Establishing surjectivity

Take an arbitrary $b \in B$. Solve $F(a) = b$ or construct a suitable a . Check that $a \in A$, then compute $F(a) = b$. To disprove surjectivity, exhibit one $b \in B$ that has no preimage.

Example 3.2: The codomain of the square map

The map $q : \mathbb{R} \rightarrow \mathbb{R}$, $q(x) = x^2$, is not surjective because -1 is not a square of a real number. The map $q : \mathbb{R} \rightarrow [0, \infty)$ with the same rule is surjective: given any $b \geq 0$, take $a = \sqrt{b} \in \mathbb{R}$ and obtain $q(a) = b$. The codomain determines whether the map is surjective.

Example 3.3: Evaluation on smooth functions

Fix $p \in U$, where $U \subseteq \mathbb{R}^n$ is open. Consider

$$\text{ev}_p : C^\infty(U) \rightarrow \mathbb{R}, \quad f \mapsto f(p).$$

It is surjective. Given $c \in \mathbb{R}$, choose the constant smooth function $f(x) = c$. Then $\text{ev}_p(f) = c$. It is not injective: the zero function and $x \mapsto x^1 - p^1$ are distinct but both take value zero at p . This example also shows why a surjective homomorphism need not be an isomorphism.

An arbitrary element of the codomain

Writing “let $b = F(a)$ ” merely selects an element already in the image. Surjectivity requires the opposite order: fix an arbitrary $b \in B$ and produce an $a \in A$ for which $F(a) = b$.

4 Bijective maps and inverses

Definition 4.1: Bijection and inverse

A map $F : A \rightarrow B$ is *bijective* if it is injective and surjective. Then every $b \in B$ has exactly one preimage $a \in A$. The *inverse map* $F^{-1} : B \rightarrow A$ assigns that unique a to b .

Proposition 4.2: Criterion for an inverse map

A map $F : A \rightarrow B$ is bijective if and only if there is a map $G : B \rightarrow A$ for which

$$G \circ F = \text{id}_A, \quad F \circ G = \text{id}_B.$$

In that case $G = F^{-1}$.

Proof. If F is bijective, its inverse satisfies both equations. Conversely, if $F(a) = F(a')$, apply G to obtain $a = a'$, so F is injective. Given $b \in B$, let $a = G(b)$. Then $F(a) = F(G(b)) = b$, so F is surjective. $\square$

One equation by itself has a more limited consequence. A *left inverse* $G \circ F = \text{id}_A$ proves that F is injective. A *right inverse* $F \circ G = \text{id}_B$ proves that F is surjective. These are distinct conditions on the compositions.

Example 4.3: The cubic map

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be $F(x) = x^3$. Given $y \in \mathbb{R}$, solving $y = x^3$ gives $x = \sqrt[3]{y}$. The map $G(y) = \sqrt[3]{y}$ satisfies $G(F(x)) = x$ and $F(G(y)) = y$. Hence F is bijective and $G = F^{-1}$.

The square map with different domains and codomains

Map $x \mapsto x^2$	Injective?	Surjective?	Reason
$\mathbb{R} \rightarrow \mathbb{R}$	no	no	1 and -1 have the same image, and negatives are missed
$\mathbb{R} \rightarrow [0, \infty)$	no	yes	each target has a square root
$[0, \infty) \rightarrow \mathbb{R}$	yes	no	unique nonnegative root, but negatives are missed
$[0, \infty) \rightarrow [0, \infty)$	yes	yes	inverse $y \mapsto \sqrt{y}$

5 Homomorphisms and isomorphisms

Injectivity and surjectivity discuss the correspondence between inputs and outputs. A *homomorphism* must also respect operations. Which operations it must preserve depends on the structures involved.

Definition 5.1: Homomorphisms

For groups $(G, \cdot)$ and $(H, \star)$, a group homomorphism $h : G \rightarrow H$ satisfies $h(ab) = h(a) \star h(b)$. For vector spaces over K , a homomorphism is a K -linear map:

$$T(u + v) = T(u) + T(v), \quad T(ru) = rT(u).$$

For unital K -algebras, a unital algebra homomorphism is K -linear and also satisfies $h(ab) = h(a)h(b)$ and $h(1_A) = 1_B$.

The homomorphism condition is checked separately for every operation required by the given structures. A map may preserve one operation and fail to preserve another. For example, a set bijection between vector spaces need not be linear: $F : \mathbb{R} \rightarrow \mathbb{R}$, $F(x) = x^3$, is bijective, but $F(1 + 1) = 8 \neq 2 = F(1) + F(1)$.

Definition 5.2: Isomorphism

An *isomorphism* of a specified kind of algebraic structure is a bijective homomorphism of that structure. A linear isomorphism is a bijective linear map. A unital algebra isomorphism is a bijective unital algebra homomorphism. The inverse of an algebraic isomorphism preserves the same operations.

Proposition 5.3: Inverse of a linear isomorphism

If $T : V \rightarrow W$ is a bijective linear map, then $T^{-1} : W \rightarrow V$ is linear.

Proof. Take arbitrary $w_1, w_2 \in W$ and $r, s \in K$. Since T is surjective, write $w_1 = T(v_1)$ and $w_2 = T(v_2)$. Then

$$T^{-1}(rw_1 + sw_2) = T^{-1}(T(rv_1 + sv_2)) = rv_1 + sv_2 = rT^{-1}(w_1) + sT^{-1}(w_2).$$

The same idea proves preservation of products for a bijective algebra homomorphism. $\square$

Conditions for the principal types of maps

Bijection of sets	Injectivity and surjectivity.
Linear isomorphism	Linearity, injectivity, and surjectivity.
Algebra isomorphism	Required algebra laws, injectivity, and surjectivity.
Diffeomorphism	Smoothness, bijectivity, and smoothness of the inverse.

The word *isomorphic* always needs an understood structure. Two sets can be bijective without being isomorphic as groups, vector spaces, or algebras. An *automorphism* is an isomorphism from a structure to itself.

6 Homeomorphisms and diffeomorphisms

Definition 6.1: Homeomorphism and diffeomorphism

A *homeomorphism* $F : X \rightarrow Y$ is a bijective continuous map whose inverse is continuous. A *diffeomorphism* $F : M \rightarrow N$ between smooth manifolds is a bijective smooth map whose inverse is smooth. For open subsets of Euclidean space, smoothness means that derivatives of all orders exist and are continuous in local coordinates.

These terms describe which structure is preserved. A diffeomorphism is also a homeomorphism, since smooth maps are continuous. Bijectivity by itself says nothing about continuity or differentiability.

Example 6.2: A smooth bijection without a smooth inverse

The map $F : \mathbb{R} \rightarrow \mathbb{R}$, $F(x) = x^3$, is smooth and bijective. Its inverse $F^{-1}(y) = \sqrt[3]{y}$ is continuous, so F is a homeomorphism. But

$$\frac{F^{-1}(h) - F^{-1}(0)}{h} = \frac{\sqrt[3]{h}}{h} = |h|^{-2/3} \quad (h \neq 0),$$

which has no finite limit at 0. Thus F^{-1} is not differentiable at 0, and F is not a diffeomorphism.

Example 6.3: The exponential diffeomorphism

The map $E : \mathbb{R} \rightarrow (0, \infty)$, $E(x) = e^x$, is smooth. The function $L : (0, \infty) \rightarrow \mathbb{R}$, $L(y) = \log y$, is smooth, and $L(E(x)) = x$, $E(L(y)) = y$. Thus E is a diffeomorphism. If its codomain were $\mathbb{R}$, it would fail to be surjective.

Local and global inverses

For a smooth map between open subsets of $\mathbb{R}^n$, an invertible derivative at one point gives a smooth inverse *near that point* by the inverse function theorem. It does not establish a global inverse. To prove a global diffeomorphism, also establish global bijectivity and smoothness of the inverse, either explicitly or by combining bijectivity with local inverses.

7 Vectors as point derivations

We now prove the identification between vectors and point derivations. Fix $p \in U \subseteq \mathbb{R}^n$, with U open. Write C_p^∞ for the algebra of germs at p of smooth real-valued functions near p .

Definition 7.1: Point derivation

A *derivation at p* is an $\mathbb{R}$ -linear map $D : C_p^\infty \rightarrow \mathbb{R}$ satisfying

$$D([fg]_p) = D([f]_p)g(p) + f(p)D([g]_p).$$

Let Der_p be the real vector space of all such derivations, with pointwise addition and scalar multiplication of maps.

For $v = (v^1, \dots, v^n) \in \mathbb{R}^n$, define

$$\Phi : \mathbb{R}^n \longrightarrow \text{Der}_p, \quad \Phi(v) = D_v, \quad D_v([f]_p) = \sum_{i=1}^n v^i \partial_i f(p).$$

We must prove that Φ is well defined and linear, then injective, then surjective. The vector v is fixed when D_v acts on a function germ $[f]_p$.

Proposition 7.2: Linearity of Φ

The displayed formula defines a point derivation for every $v \in \mathbb{R}^n$, and Φ is $\mathbb{R}$ -linear.

Proof. If two representatives have the same germ at p , they agree on a neighbourhood of p . Their partial derivatives at p therefore agree, so $D_v([f]_p)$ is independent of the representative. Linearity in $[f]_p$ follows from linearity of partial derivatives. The product rule gives

$$D_v([fg]_p) = \sum_i v^i (\partial_i f(p)g(p) + f(p)\partial_i g(p)) = D_v([f]_p)g(p) + f(p)D_v([g]_p).$$

Finally, for $v, w \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$, evaluating on any $[f]_p$ shows $D_{av+bw}([f]_p) = aD_v([f]_p) + bD_w([f]_p)$. Hence $\Phi(av + bw) = a\Phi(v) + b\Phi(w)$. $\square$

Proposition 7.3: Injectivity of Φ

The map Φ is injective.

Proof. Since Φ is linear, it suffices to show that its kernel is zero. Fix $v \in \mathbb{R}^n$ and suppose $\Phi(v) = D_v = 0$, the zero derivation. The equality $D_v = 0$ means that $D_v([f]_p) = 0$ for *every* germ, so we may choose a useful one. For each j , let x^j denote the j th coordinate function. Then $\partial_i x^j(p) = \delta_i^j$, and

$$0 = D_v([x^j]_p) = \sum_{i=1}^n v^i \delta_i^j = v^j.$$

Every component of v is zero. Thus $v = 0$, so $\ker \Phi = \{0\}$. $\square$

The coordinate functions need not represent arbitrary germs. The hypothesis $D_v = 0$ asserts that the operator vanishes on every germ. Evaluation on the coordinate germs isolates the components of the fixed vector v .

Proposition 7.4: Surjectivity of Φ

Every derivation $D \in \text{Der}_p$ equals D_v for a unique $v \in \mathbb{R}^n$. Hence Φ is surjective.

Proof. Let $D \in \text{Der}_p$ be arbitrary. Define a vector v by

$$v^i = D([x^i]_p), \quad i = 1, \dots, n.$$

We must show $D([f]_p) = D_v([f]_p)$ for *every* smooth germ. The Leibniz rule gives $D([1]_p) = D([1]_p[1]_p) = 2D([1]_p)$, so $D([1]_p) = 0$. By $\mathbb{R}$ -linearity, D kills every constant germ.

Choose a small convex ball around p on which f is defined. On that ball,

$$f(x) - f(p) = \sum_{i=1}^n (x^i - p^i) g_i(x), \quad g_i(x) = \int_0^1 \partial_i f(p + t(x - p)) dt.$$

This identity follows by applying the fundamental theorem of calculus to $t \mapsto f(p + t(x - p))$. Each g_i is smooth near p , and $g_i(p) = \partial_i f(p)$. Passing to germs and applying D gives

$$\begin{aligned} D([f]_p) &= \sum_i D([x^i - p^i]_p [g_i]_p) \\ &= \sum_i (D([x^i - p^i]_p) g_i(p) + (x^i(p) - p^i) D([g_i]_p)) \\ &= \sum_i D([x^i]_p) \partial_i f(p) = \sum_i v^i \partial_i f(p) = D_v([f]_p). \end{aligned}$$

Thus $D = D_v = \Phi(v)$ as maps on all germs. Uniqueness of v follows from injectivity. $\square$

The injectivity argument recovers a vector from the values of its derivation on coordinate functions. The surjectivity argument constructs a vector from an arbitrary derivation and proves equality of the resulting operators on every germ.

The three propositions prove that $\Phi : \mathbb{R}^n \rightarrow \text{Der}_p$ is a linear isomorphism. Its inverse is concrete:

$$\Phi^{-1}(D) = (D([x^1]_p), \dots, D([x^n]_p)).$$

8 Proof methods and exercises

8.1 Summary of the proof criteria

Standard proof steps

1. State the domain and codomain. If inputs are equivalence classes, check well-definedness first.
2. For injectivity, take two inputs with equal images, or take an element of the kernel when the map is linear.
3. For surjectivity, take an arbitrary target and construct a preimage.
4. For a homomorphism, verify the required operations. For a diffeomorphism, verify smoothness in both directions.

Insufficient arguments

- Checking $F(a) \neq F(a')$ for one pair does not prove injectivity. The claim is about *all* distinct pairs.
- Producing one output $F(a)$ does not prove surjectivity. The target must be arbitrary in the stated codomain.
- A bijective map of sets is not automatically an isomorphism of vector spaces or a diffeomorphism. Check the relevant structure.

For smooth vector fields, the following pointwise operations will be used. If X is a smooth vector field and f is smooth, then $(Xf)(p) = X_p(f) = \sum_i X^i(p) \partial_i f(p)$ defines a smooth *function*. For a smooth function h , the field hX has value $(hX)_p = h(p)X_p$.

The following exercises apply the preceding criteria. In each case, the domain and codomain form part of the statement to be proved.

8.2 Exercises

1. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be $T(x, y) = (x + y, x - y)$. Prove that T is a linear isomorphism and find its inverse.
2. Let $P : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be $P(x, y, z) = (x + z, y)$. Determine whether it is injective and surjective. Give witnesses.
3. Consider $\text{ev}_p : C^\infty(U) \rightarrow \mathbb{R}$. Is it an isomorphism of real vector spaces? Explain.

4. Let $F : \mathbb{R} \rightarrow \mathbb{R}$, $F(x) = x^3$. Which of “bijection”, “homeomorphism”, and “diffeomorphism” apply?
5. Suppose $F : A \rightarrow B$ has a left inverse $G : B \rightarrow A$. Prove injectivity directly from $F(a) = F(a')$. Does this alone prove surjectivity? Give an example.
6. Fix $p \in U$. Show that $J : C_p^\infty \rightarrow \mathbb{R}$ given by $J([f]_p) = \partial_1 f(p)$ is well defined, linear, and surjective. Is it injective?
7. For $n \geq 2$, consider $Q : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $k \mapsto [k]$. Prove that it is an additive group homomorphism and surjective. Is it injective? Is it an isomorphism of groups?
8. For a smooth vector field X and smooth functions f, h on U , prove $(hX)f = h(Xf)$ by evaluating at an arbitrary point. Which of hX and Xf is a vector field?

8.3 Solutions

1. Componentwise calculation gives $T(a(x, y) + b(u, v)) = aT(x, y) + bT(u, v)$, so T is linear. Given $(s, t) \in \mathbb{R}^2$, solve $s = x + y$, $t = x - y$. The unique solution is $x = (s + t)/2$, $y = (s - t)/2$. Thus $T^{-1}(s, t) = ((s + t)/2, (s - t)/2)$, and the formulas compose to the identity in both orders. Therefore T is a linear isomorphism.
2. The nonzero vector $(1, 0, -1)$ lies in $\ker P$, so P is not injective. Given $(a, b) \in \mathbb{R}^2$, take $(x, y, z) = (a, b, 0)$. Its image is (a, b) . Hence P is surjective.
3. Evaluation is linear and surjective by constant functions. It is not injective: $f(x) = x^1 - p^1$ is a nonzero smooth function with $f(p) = 0$. Therefore it is not a vector-space isomorphism.
4. The cubic is bijective with continuous inverse $y \mapsto \sqrt[3]{y}$, so it is a homeomorphism. Its inverse is not differentiable at 0, so it is not a diffeomorphism.
5. If $F(a) = F(a')$, then $a = G(F(a)) = G(F(a')) = a'$. This proves injectivity. The inclusion $F : \{0\} \rightarrow \{0, 1\}$ has a left inverse $G : \{0, 1\} \rightarrow \{0\}$, but F misses 1, so a left inverse alone does not prove surjectivity.
6. Equal germs agree on a neighbourhood of p , so their first partial derivatives at p agree. This proves well-definedness. The map is linear by the derivative rules. Given $c \in \mathbb{R}$, choose $f(x) = cx^1$. Then $J([f]_p) = c$. The nonzero constant germ $[1]_p$ lies in the kernel, so J is not injective.
7. Since $[k + \ell] = [k] + [\ell]$, the map is a group homomorphism. Given a residue class $[r]$, the integer r maps to it, so Q is surjective. But $Q(0) = Q(n) = [0]$ and $0 \neq n$. It is not injective and therefore not a group isomorphism.
8. At any $p \in U$, the field hX has value $(hX)_p = h(p)X_p$. Hence $((hX)f)(p) = (hX)_p(f) = h(p)X_p(f) = (h(Xf))(p)$. Equality at every p proves $(hX)f = h(Xf)$ as functions. The object hX is a vector field. The object Xf is a smooth function.